



Transparent Rectenna Array for Space Applications



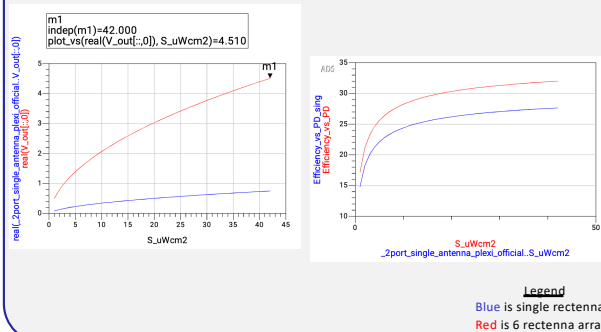
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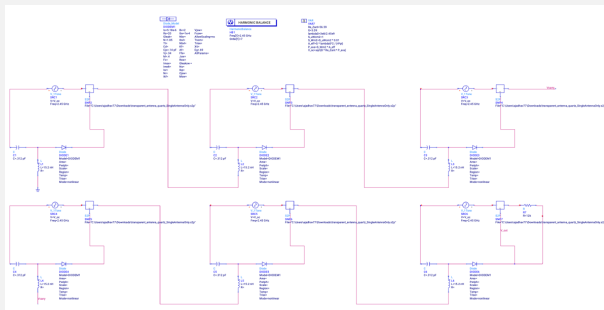
Motivation

- Spacecrafts demand lightweight, non-intrusive power solutions for low-consumption electronics, where every gram and square centimeter counts.
- Solutions must integrate seamlessly into existing surfaces, viewports, optical windows, and sensor covers, without obstructing visibility.
- **Goal:** Develop and model a transparent rectenna array module on a quartz substrate to harvest low ambient RF energy and transform it into usable DC power.

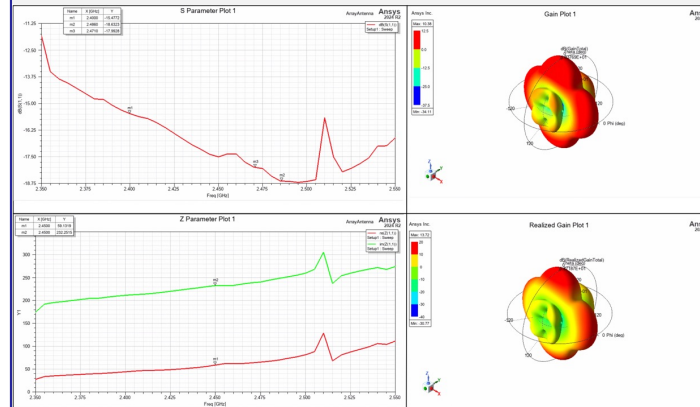
ADS Results



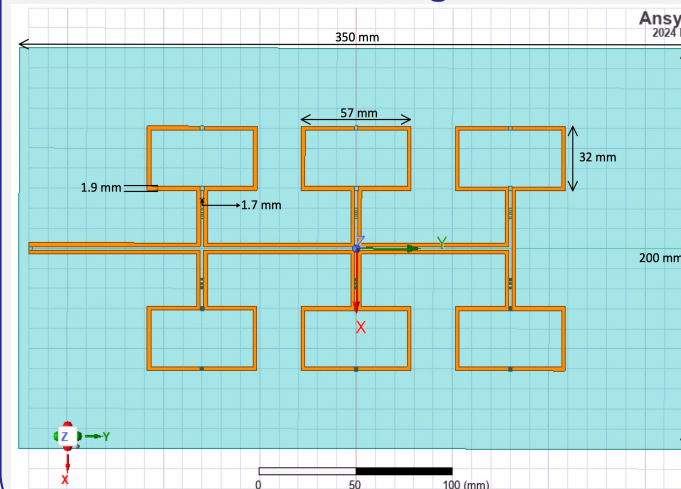
ADS Schematic



HFSS Results



HFSS Design



Conclusion

- **Substrate Impact on Efficiency:** The choice of substrate material significantly impacts RF-to-DC conversion. The simulated single-element quartz rectenna matched the efficiency of the plexiglass substrate while making it feasible for extraterrestrial deployments.
- **Array Performance Enhancement:** Expanding the design to a 6-element quartz array significantly improved energy harvesting capabilities, achieving a simulated efficiency of 34%. At a power density of $5 \mu\text{W}/\text{cm}^2$, the quartz array produced an output voltage of approximately 4V, demonstrating highly effective power multiplication.
- **Voltage Output:** The single-element quartz rectenna produced a simulated output voltage of $\sim 0.68\text{V}$ at $40 \mu\text{W}/\text{cm}^2$. While slightly lower than the theoretical expectation of $\sim 0.7\text{V}$ at $40 \mu\text{W}/\text{cm}^2$, it successfully demonstrates viable low-ambient RF energy harvesting.
- **Material Properties:** The matched performance of the quartz design is directly attributed to its lower dielectric loss tangent (0.0002) compared to plexiglass (0.001), which minimizes internal energy dissipation within the substrate and maximizes simulated power delivery to the rectifying diode.

Next Steps

- Integrate and test advanced multi-stage rectification circuits (e.g., Mandal charge pumps) with the quartz substrate antenna to evaluate improvements in DC output voltage and RF-to-DC efficiency under controlled RF power conditions.
- Fabricate and characterize quartz substrate antennas using alternative transparent conductive materials (e.g., ITO) to assess the trade-off between optical transparency and RF performance compared to coppered designs.

References

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